

## exp 2

```
import nltk
import matplotlib.pyplot as plt
import random
```

```
import nltk
nltk.download('twitter_samples')
```

```
[nltk_data] Downloading package twitter_samples to /root/nltk_data...
[nltk_data]   Package twitter_samples is already up-to-date!
True
```

```
import nltk
from nltk.corpus import twitter_samples
```

```
twitter_samples
```

```
<TwitterCorpusReader in '/root/nltk_data/corpora/twitter_samples'>
```

```
tweets = twitter_samples.strings()
type(tweets)
```

```
list
```

```
len(tweets)
```

```
30000
```

```
import pandas as pd
all_tweets=pd.DataFrame(tweets,columns=['Tweets'])
all_tweets.head()
```

|   | Tweets                                            |
|---|---------------------------------------------------|
| 0 | hopeless for tmr :(                               |
| 1 | Everything in the kids section of IKEA is so c... |
| 2 | @Hegelbon That heart sliding into the waste ba... |
| 3 | "@ketchBurning: I hate Japanese call him "bani... |
| 4 | Dang starting next week I have "work" :(          |

```
import pandas as pd
all_tweets=pd.DataFrame(tweets,columns=['Tweets'])
all_tweets.head()
```

|   | Tweets                                            |
|---|---------------------------------------------------|
| 0 | hopeless for tmr :(                               |
| 1 | Everything in the kids section of IKEA is so c... |
| 2 | @Hegelbon That heart sliding into the waste ba... |
| 3 | "@ketchBurning: I hate Japanese call him "bani... |
| 4 | Dang starting next week I have "work" :(          |

```
print("Length of the first tweet: ", len(tweets[2]))
```

```
Length of the first tweet: 54
```

```
twitter_samples.fileids()
```

```
['negative_tweets.json', 'positive_tweets.json', 'tweets.20150430-223406.json']
```

```
positive_tweets = twitter_samples.strings('positive_tweets.json')
negative_tweets = twitter_samples.strings('negative_tweets.json')
```

```
negative_tweets[1]
```

```
'Everything in the kids section of IKEA is so cute. Shame I'm nearly 19 in 2 months :('
```

```
import pandas as pd
positive_tweets=pd.DataFrame(positive_tweets,columns=['Tweets'])
positive_tweets['Sentiments']='positive'
```

```
positive_tweets.head()
```

|   | Tweets                                            | Sentiments |
|---|---------------------------------------------------|------------|
| 0 | #FollowFriday @France_Inte @PKuchly57 @Milipol... | positive   |
| 1 | @Lamb2ja Hey James! How odd :/ Please call our... | positive   |
| 2 | @DespiteOfficial we had a listen last night :)... | positive   |
| 3 | @97sides CONGRATS :)                              | positive   |
| 4 | yeaaaaah yippppy!!! my acct verified rqst has...  | positive   |

```
negative_tweets=pd.DataFrame(negative_tweets,columns=['Tweets'])
negative_tweets['Sentiments']='negative'
```

```
negative_tweets.head()
```

|   | Tweets                                            | Sentiments |
|---|---------------------------------------------------|------------|
| 0 | hopeless for tmr :(                               | negative   |
| 1 | Everything in the kids section of IKEA is so c... | negative   |
| 2 | @Hegelbon That heart sliding into the waste ba... | negative   |
| 3 | "@ketchBurning: I hate Japanese call him "bani... | negative   |
| 4 | Dang starting next week I have "work" :(          | negative   |

```
tweets=pd.concat([positive_tweets, negative_tweets], ignore_index=True)
tweets = tweets.sample(frac=1).reset_index(drop=True)
tweets
```

|      | Tweets                                            | Sentiments |
|------|---------------------------------------------------|------------|
| 0    | @nidhi_budha Aja besi chati phulani swasa baha... | positive   |
| 1    | We are the one and only approve and recognized... | positive   |
| 2    | I still have over 6 hours of work left omg :(     | negative   |
| 3    | @DeviousLiz awh :-(💙 but you have to stay💛        | negative   |
| 4    | @realiseagency you too :)                         | positive   |
| ...  | ...                                               | ...        |
| 9995 | but wait! it's the 2nd year soon :) they get t... | positive   |
| 9996 | @Barrastinian @chunkymark they would call it t... | positive   |
| 9997 | @JackJackJohnson ended up happy. :)))             | positive   |
| 9998 | I said gn along time ago my ass can't fall asl... | negative   |
| 9999 | @LigiaMonroy don't forget to dream about me :)    | positive   |

10000 rows × 2 columns

```
print("number of positive tweets: ", len(positive_tweets))
print("number of negative tweets: ", len(negative_tweets))
```

```
number of positive tweets: 5000
number of negative tweets: 5000
```

```
tweets[tweets['Sentiments']=="negative"].value_counts().sum()
tweets[tweets['Sentiments']=="positive"].value_counts().sum()
```

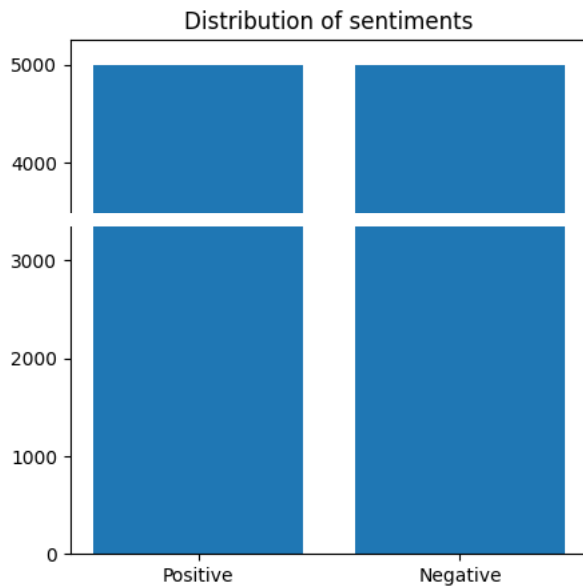
```
np.int64(5000)
```

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(5,5))
```

```
labels = 'Positive', 'Negative'
```

```
sizes = [tweets[tweets['Sentiments']=="positive"].value_counts().sum(), tweets[tweets['Sentiments']=="negative"].value_counts().sum()]
```

```
plt.bar(labels,sizes)
plt.title('Distribution of sentiments')
plt.show()
```



```
#preprocessing
tweet = positive_tweets.iloc[5]
print(tweet)
```

```
Tweets      @BhaktisBanter @PallaviRuhail This one is irre...
Sentiments                                positive
Name: 5, dtype: object
```

```
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
True
```

```
tweet1 = all_tweets.iloc[2005]
tweet1["Tweets"]
```

```
'Delph??! 20 minutes, off on his debut :('
```

```
import re
import string
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.tokenize import TweetTokenizer
```

```
tweet1= tweets["Tweets"][2005]
tweet1
```

```
'@TnmntTiles Our tiles, while a nice green, are boring - but I love the original glass door the guy downstairs has :) http://t.co/fxgpk8hNi'
```

```
tweet2 = re.sub(r'^RT[\s]+', '', tweet1)

tweet2 = re.sub(r'https?:\/\/[^\s\n\r]+', '', tweet2)

tweet2 = re.sub(r'#', '', tweet2)
tweet2 = re.sub(r'@\w+', '', tweet2)
tweet2 = re.sub(r'[:]|(:)]', '', tweet2)
print(tweet1)
print("\n Cleaned tweet: \n",tweet2)
```

```
@TnmntTiles Our tiles, while a nice green, are boring - but I love the original glass door the guy downstairs has :) http://t.co/fxgpk8hNi

Cleaned tweet:
Our tiles, while a nice green, are boring - but I love the original glass door the guy downstairs has
```

```
tokenizer = TweetTokenizer(preserve_case=False, strip_handles=True, reduce_len=True)
```

```
tweet_tokens = tokenizer.tokenize(tweet2)
```

```
print()
print("Tokenized string:\n")
print(tweet_tokens)
```

Tokenized string:

```
['our', 'tiles', ',', 'while', 'a', 'nice', 'green', ',', 'are', 'boring', '-', 'but', 'i', 'love', 'the', 'original', 'glass', 'door', 't
```

```
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
True
```

```
stopwords_english = stopwords.words('english')
print('stop words\n')
print(stopwords_english)
print('\npunctuation\n')
print(string.punctuation)
```

stop words

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', 'and', 'any', 'are', 'aren', "aren't", 'as', 'at', 'be', 'b
```

punctuation

```
!"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~
```

```
print(tweet_tokens)
print()
tewwts_clean = []
for word in tweet_tokens:
    if (word not in stopwords_english and word not in string.punctuation):
        tewwts_clean.append(word)
print("removed stop words and punctuation:")
print(tewwts_clean)
```

```
['our', 'tiles', ',', 'while', 'a', 'nice', 'green', ',', 'are', 'boring', '-', 'but', 'i', 'love', 'the', 'original', 'glass', 'door', 't
```

removed stop words and punctuation:

```
['tiles', 'nice', 'green', 'boring', 'love', 'original', 'glass', 'door', 'guy', 'downstairs']
```

```
def process_tweets(tweet):
    tokenizer = TweetTokenizer(preserve_case=False, strip_handles=True, reduce_len=True)

    tweet = re.sub(r'^RT[\s]+', '', tweet)
    tweet = re.sub(r'https?:\/\/[^\s\r\n]+', '', tweet)
    tweet = re.sub(r'#', '', tweet)

    tweet_tokens = tokenizer.tokenize(tweet)
    stopwords_english = stopwords.words('english')
    tweets_clean = []
    for word in tweet_tokens:
        if (word not in stopwords_english and word not in string.punctuation):
            tweets_clean.append(word)
    return tweets_clean
```

```
import nltk
import pandas as pd
import re
import string
from nltk.corpus import twitter_samples, stopwords
from nltk.tokenize import TweetTokenizer

nltk.download('twitter_samples')
nltk.download('stopwords')

def process_tweets(tweet):
    tokenizer = TweetTokenizer(preserve_case=False, strip_handles=True, reduce_len=True)
```

```
    tweet = re.sub(r'^RT[\s]+', '', tweet)
```

```

tweet = re.sub(r'RT[\s]+', '', tweet)
tweet = re.sub(r'https?:\/\/[^\s\n\r]+', '', tweet)
tweet = re.sub(r'#', '', tweet)

tweet_tokens = tokenizer.tokenize(tweet)
stopwords_english = stopwords.words('english')
tweets_clean = []
for word in tweet_tokens:
    if (word not in stopwords_english and word not in string.punctuation):
        tweets_clean.append(word)
return tweets_clean

tweets = twitter_samples.strings()
all_tweets = pd.DataFrame(tweets, columns=['Tweets'])

processed_tweet = all_tweets['Tweets'].apply(process_tweets)
all_tweets['processed_tweets'] = processed_tweet
pd.set_option('display.max_colwidth', None)
all_tweets

```

```

[nltk_data] Downloading package twitter_samples to /root/nltk_data...
[nltk_data] Package twitter_samples is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.

```

|       | Tweets                                                                                                                                         | processed_tweets                                                                                     |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 0     | hopeless for tmr :(                                                                                                                            | [hopeless, tmr, :(]                                                                                  |
| 1     | Everything in the kids section of IKEA is so cute. Shame I'm nearly 19 in 2 months :(                                                          | [everything, kids, section, ikea, cute, shame, nearly, 19, 2, months, :(]                            |
| 2     | @Hegelbon That heart sliding into the waste basket :(                                                                                          | [heart, sliding, waste, basket, :(]                                                                  |
| 3     | "@ketchBurning: I hate Japanese call him "bani" :( ("n\nMe too                                                                                 | [", hate, japanese, call, bani, :(, :(", "                                                           |
| 4     | Dang starting next week I have "work" :(                                                                                                       | [dang, starting, next, week, work, :(]                                                               |
| ...   | ...                                                                                                                                            | ...                                                                                                  |
| 29995 | RT @UKLabour: @Ed_Miliband: we're not going to do a deal with the SNP, there will be no coalition with the SNP. #bbcqt\nhttps://t.co/n65hTe... | [going, deal, snp, coalition, snp, bbcqt]                                                            |
| 29996 | RT @DisabledScot: @blairmcdougall @ScotlandTonight @bbcqt If Ed thinks Scots will suddenly rush to vote Labour, we'll vote SNP, Labour just... | [ed, thinks, scots, suddenly, rush, vote, labour, vote, snp, labour, ...]                            |
| 29997 | RT @Staircase2: @VividRicky exactly but that alleged commute was made up by Farage as was his 'discomfort' in allegedly 'not hearing Englis... | [exactly, alleged, commute, made, farage, discomfort, allegedly, hearing, englis, ...]               |
| 29998 | Actually agreed with %95 of what farage was sayin rhen#voteukip                                                                                | [actually, agreed, 95, farage, sayin, rhenvoteukip]                                                  |
| 29999 | RT @PeatWorrier: "A vote for the SNP is. Er. A vote for another referendum we insist Holyrood has no legal authority to legislate for" http... | [vote, snp, er, vote, another, referendum, insist, holyrood, legal, authority, legislate, http, ...] |

## EXP 3

```

# Import required libraries
import re
import string
import nltk
!pip install emoji
import emoji

# Download required NLTK resources
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer

```

```

Collecting emoji
  Downloading emoji-2.15.0-py3-none-any.whl.metadata (5.7 kB)
  Downloading emoji-2.15.0-py3-none-any.whl (608 kB)
----- 608.4/608.4 kB 29.9 MB/s eta 0:00:00
Installing collected packages: emoji
Successfully installed emoji-2.15.0
[nltk_data] Downloading package punkt to /root/nltk_data...

```